Dancing with Nature : Transforming Dai Dance Movements into Interactive Music and Visuals

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***Abstract*—Currently, digital experiences of traditional dance are often limited to passive viewing, making it difficult for users to understand cultural significance through physical participation. To address this issue, this study proposes a multimodal interaction system based on embodied cognition theory, using an ethnic dance from the Dai ethnic group as a case study. The system features a multilayered structure and translates user movements into real-time musical and visual experiences. The underlying "body structure control layer" captures motion data via PoseNet models and pressure sensors. It maps these inputs to musical melodies, rhythms, and basic visual textures in a Max/MSP environment, ensuring low-latency feedback. The higher-level "motion morpheme recognition layer" identifies culturally significant movement patterns such as "Three Bends" and "Peacock Hand", triggering Dai tribe-specific melodic phrases and displaying cultural visual symbols within TouchDesigner. Experimental results show that the system provides an immersive cultural experience for ordinary users, effectively enhancing their appreciation of dance aesthetic and cultural symbolism.**

Keywords—Motion-to-Music, Dai Dance, Cultural Experience, Embodied Interaction Multimodal System, Digital Cultural Creativity

I. INTRODUCTION

The high-speed innovation of digital technology and interaction design is remolding the ways of expressing and experiencing the culture and creative industries. However, the traditional dance experiences still depend predominantly on viewing or watching videos in real time. Without actual participation, the general users cannot create an organic relationship between action patterns and cultural images. Based on embodied design theory, the body is not merely a way of executing movement, but an essential way of cultural perception [1][2]. By converting the movement of the body into dynamic music and visual feedback in real time, the users can sense the aesthetic value of the Dai dance during interaction[3][4], broadening cultural participation and perception [5][6].

The "Dance with Nature" multimedia interactive system employs a two-layer structure to transform users' body postures, hand movements, and rhythm patterns into real-time music and visual responses: the lower layer uses the body structure to provide low-latency data feedback, while the upper layer activates Dai dance's unique music phrases and graphics by detecting movement changes. This system uses PoseNet to capture the key points of the body and arms, pressure sensors track the center of gravity, and transmits the data stream to Max/MSP and TouchDesigner (abbreviated as TD hereafter). This simultaneously gives rise to music with the Dai ethnic style and natural visual image.

The worth of the study is in that it can help to: (1) increase the depth of the users' cultural experience. The non-professional users take part in the cultural exchanges by means of body movement and directly experience the activity-culture image connection. (2) Encourage interdisciplinary innovation: The combination of dance, music, visual design, and human-computer interaction technology offers available digital models of interaction for the cultural and creative sectors. (3) Open up new areas of cultural dissemination: By means of multimodal interactive experience, offer practical assistance in digital exhibitions, immersive performance, and cultural education activities, to cultivate the digital preservation and industrial development of national culture.

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II. RELATED WORK

A. Ethnic Dance Characteristics and Digitalization

Ethnic dances constitute a comprehensive art form that embodies regional culture and natural philosophy. Taking the Dai dance as an example, its movements are fluid, its dance processed rhythmically with the music to form rhythmic breaths. And dance elements combining natural imagery such as peacocks and flowing water, show the aesthetic concept of "harmony between humanity and heaven" [7][8][9][10]. Digital technology offers new avenues for preserving and disseminating dance. At the motion computation level, generative models and spatio-temporal feature analysis enable the recognition model and semantic understanding of stylized movements specific to particular dance genres [11][12]. Pat Pataranutaporn et al. proposed a human-machine collaborative choreography paradigm, where generative virtual characters respond to and co-create with dancers in real time [13]. Su et al. employed a multimodal interactive approach to create a digital educational game for the preservation of ethnic dances, thereby enhancing players' engagement and experience during the dance process. [14].

B. Motion-to-Music and Motion-to-Visuals

Research on motion-generated music has progressed from direct or generative mapping toward embodied sonification. Early work primarily relied on direct control of sound through body or gesture parameters, such as the EyesWeb platform and the "Modular Music Object" [15][16]. Subsequent studies introduced probabilistic and generative mappings [17], while systems such as Brown and Sorensen's Live Coding approach and Françoise et al.'s CO/DA framework enabled human–computer collaboration and dance-based improvisation [18][19]. Theories of embodied music cognition [20], body-sonification experiments, and analyses of dance rhythmic structure further reveal the intrinsic coupling between movement, rhythm, and perception[21][22].

Research on motion-driven visualization has similarly achieved real-time conversion from gesture and posture to abstract animation and audio–visual synchronization [23][24], supported by evidence of stable psychological correspondences between auditory parameters and visual attributes[25]. The ANGIE framework extends these capabilities by introducing vector-quantized motion extraction and gesture-specific GPT models for converting speech audio into photorealistic gesture video sequences [26]. However, most existing systems remain oriented toward generalized technical mappings and leave the culturally specific body vocabulary and symbolic systems of folk dance insufficiently integrated, limiting their capacity for cultural narrative and emotional expression.

C. *Multimodal Interaction and Digital Cultural Experience*

Multimodal interaction shifts users from passive cultural observers to active participants by integrating movement, sound, and visual channels, generating richer cognitive and affective engagement and supporting embodied cultural understanding [27]. Merleau-Ponty's phenomenology of embodiment establishes the foundational view that perception and action are inseparable, positioning the body as the primary medium through which meaning is constituted [28]. Building on this grounding, Benford and Giannachi's framework of performance, immersion, and experiential trajectories further explains how cultural experience arises through the interplay between technological mediation and situated cultural practice [29].

Systems such as the MuMIA interface demonstrate this principle, using eye-tracking and voice communication to enhance users' contextual understanding and participation in cultural exhibitions [30]. Research in embodied interaction shows that multisensory tactile interfaces transform visitors from passive information recipients into co-constructors of meaning, enabling abstract cultural concepts to be perceived through collaborative multimedia and tangible experience [31][32]. As Dourish argues, the significance of embodied interaction lies not only in the technical system, but also in the cultural practices and interpretive processes embedded within it [3]. This perspective underscores a recurring limitation in existing work: the insufficient integration of cultural semantics.

III. Methodology & System Design

A. *Overall System Architecture*

This paper proposes an interactive system based on embodied cognition, integrating dance, music, and vision. This theory emphasizes that physical movement is the result of perception and the main means of constructing consciousness. Dance, as an art form that relies on body movements, is a necessary condition for understanding national aesthetics and their logic. Therefore, this system takes advantage of the flow of consciousness, providing a closed-loop interaction among "body, music, and culture" through its real-time perception and multimodal feedback.

The system adopts a two-layer architecture design, namely the "body structure control layer" and the "action morpheme recognition layer", to achieve real-time and multi-modal mapping from dance movements to music and visuals.

The "Body structure control layer" collects basic signals and parameter mappings, ensuring high feedback response speed and low latency between movements and music graphics. It captures the physical features of dancers such as their movements, speeds and pressures in real time, converts them into basic data of music rhythms, melodies and dynamic graphics, and provides users with immediate interactive feedback. The "action morpheme recognition layer" conducts semantic analysis on specific action combinations (such as "Three Bends", "Peacock Hand") and abstracts them into computable ones. The former shows melodic phrases that conform to the logic of ethnic music, while the latter generates beautiful graphics with cultural images, enhancing the aesthetic quality and cultural depth of the system.

The whole design insists on multimodal interaction logic. Simultaneously send motion information to two sensory channels. The result is a deeply integrated experience of movement, hearing, and vision. The system has four core modules: motion capture, data mapping, sound synthesis, and image generation, forming a complete real-time interaction cycle.

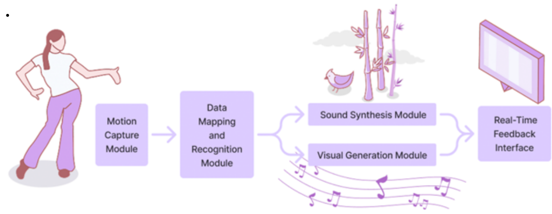

Fig. 1. Multimodal interaction logic.

B. *Motion Capture and Sensor Design*

The system integrates computer vision and physical sensing. Using the ml5.js implementation of PoseNet (640×480 px, 60 Hz), it extracts 17 body key points for whole-body motion, while Hand pose captures fine hand articulations at the same frame rate. A foot-mounted pressure sensor, calibrated through zero-load, partial-load, and full-load stages, provides continuous measurements of weight distribution and center-of-gravity shifts. All visual and pressure data are streamed to the system and jointly form a unified body-movement representation, which is subsequently mapped to music and visual outputs.

C. *Action-to-Music Mapping Rules*

The module of the music generation holds the double-layered logic of the structure of "body structure control + action morpheme triggering." Based on embodied cognition theory, the movement of the body is regarded as the intrinsic motivator of musical generation.

- Rhythm Mapping

  In this system, a motion morpheme refers to the smallest meaningful unit of bodily movement that can be computationally detected and mapped to sound. Accordingly, the system receives pressure information from the feet through pressure sensors and translates these motion morphemes into rhythmic intensities. High pressure leads to strong beats, low pressure leads to weak beats, and jumping off the ground (zero pressure) generates rests. These mappings produce a rhythm that is synchronised with the body's expressive dynamics. The "Max MSP" drum set employs the 2/4 rhythmic pattern commonly found in Dai dance and incorporates culturally specific percussion timbres—such as the elephant-foot drum and xylophone—to maintain stylistic compatibility. This rhythmic design draws from the embodied breathing structure characteristic of Dai dance: "heavy beats sink, light beats rise." In this context, cultural imagery is defined as the symbolic associations and aesthetic metaphors evoked by these rhythmic expressions, linking bodily actions to culturally grounded sonic interpretations. The system operationalises this principle through the following mapping logic:

TABLE I. RHYTHM MAPPING LOGIC

Body Expression	System Feedback	Cultural Imagery
Foot pressing down, body sinking	Trigger: strong beat	The force of nature and ritual solemnity.
	Sound: elephant-foot drum, gong	
Foot relaxing, body extending upward	Trigger: weak beat	Elegance, fluidity, and the aesthetic of natural movement.
	Sound: guqin, wooden fish	
Jumping or leaving the ground (pressure = 0)	Trigger: rest symbol	Enhancing rhythmic breathing and contrast within dynamic flow.
	Control logic: pressure signal deactivated	

Foot pressure controls note velocity using:

$$velocity_i = g(S_i) = k \cdot S_i \quad (1)$$

Here, k is the dynamic amplification coefficient, which is used to control the dynamic range of the note.

Establishes a direct relationship among the body's center of gravity, rhythm and rhythm patterns, and draws on the essence of the concept that "the body is rhythm". The movement of the user's center of gravity shapes the time structure of the music, forming a closed loop between the dance rhythm and the breathing of the music.

- Melody Mapping

The melody is based on the Dai pentatonic scale mode, mainly the Yu mode, mapping the spatial movements of key trunk nodes onto the melodic line. This system changes the range and pitch of the melody by observing the displacement of nodes on the X-axis and Y-axis of the two-dimensional plane in real time.

Horizontal body expansion is normalized and mapped to melodic range: when the body contracts, the melody oscillates within a narrow 3-semitone span characteristic of traditional Dai "small-range wave" contours; when the body fully expands, the melodic span increases up to approximately 8 semitones, producing a more spatial and outward musical gesture. Vertical movement directly shapes pitch and loudness. The height of the highest body keypoint is mapped to a one-octave pitch span (C4–C5). Loudness increases correspondingly from -18 dB to -6 dB, allowing higher gestures to produce brighter and more projected timbres, while lower gestures generate softer, grounded sonorities.

x_i and y_i denote the lateral and longitudinal displacements of the torso center point at frame i, respectively. a, b, and c are coefficients defined according to the pitch range of the Dai pentatonic scale.

$$pitch_i = p_0 + \alpha \cdot y_i + \beta \cdot (x_i - \mu_x) \quad (2)$$

In Max/MSP, the melodies are generated in real time and use Dai ethnic instruments such as the Bawu flute and the Hulusi. This enables melody directly generated by the movements to respond in real time and present unique sound of Dai people, embody the idea that "the body is the melody".

- Morpheme Mapping

The system defines a set of cultural symbol morphemes, primarily involving the "Three Bends" posture and a series of associated gestures.

In Dai dance, the "Three Bends" refers to the coordinated bending of the neck, waist, knees, and other joints, forming an elegant S-shaped contour that evokes natural metaphors such as flowing water or swaying bamboo. The system employs joint-angle threshold detection for identifying the "Three Bends" posture. For the lower limbs, when the angle formed by the knee, hip, and ankle joints falls within 80–160°, the posture is recognized as a three-bend configuration. For the upper limbs, a three-bend posture is identified when the angle formed by the wrist, elbow, and shoulder joints is within the same 80–160° range. Once a three-bend posture is detected in either the upper or lower limbs, the system triggers a pre-programmed wave-like musical phrase characteristic of Dai musical aesthetics. These wave-like phrases typically follow an undulating contour, oscillating repeatedly across two to three scale registers, rising and falling continuously like rippling water.

TABLE II. MORPHEME MAPPING LOGIC

Body Expression	System Feedback	Cultural Imagery
Lower-limb "Three Bends" 	Trigger: Lower part of the body synchronized torso inclination. Calculate the angles of the hip, knee and ankle joints.	The unity of strength and grace.
	Sound: preset fluctuating phrases	
Upper-limb "Three Bends" 	Trigger: coordinated curvature of wrist, elbow, and shoulder joints	Elegance, lightness, and vitality.
	Sound: preset fluctuating phrases	

In addition to the "Three Bends" posture, the system also maps specific gesture morphemes to corresponding natural sound responses. Using p5.js, the system performs precise recognition of three culturally meaningful gestures—the Peacock Gesture, the Leaf Gesture, and the Fish Gesture. Hand gesture recognition triggers pre-designed musical motifs through the following discrete function:

$$trigger_i = h(G_i), \quad (3)$$

where G_i is the detected gesture type of frame i.

The incorporation of these natural sonic elements transforms music generation into an extension of cultural semantics, enriching the expressive landscape of the composition and extending beyond simple or direct parameter-based control.

TABLE III. HAND GESTURE MAPPING

Body Expression	System Feedback	Cultural Imagery
Peacock Gesture 	Trigger: The thumb and index finger contact.	beauty, vitality, and auspiciousness
	Sound: bird chirping.	

Body Expression	System Feedback	Cultural Imagery
Leaf Gesture 	Trigger: The thumb is close to the index finger.	purity, resilience, and serenity
	Sound: bamboo forest ambience.	
Fish Gesture 	Trigger: bilateral hand overlap.	fluidity, adaptability, harmony
	Sound: flowing water.	

D. Action-to-Visual Mapping Rules

The visual generation module designed on the TD platform centers on the natural images and cultural symbols of Dai ethnic dances. It converts the data stream of object movement into a dynamic visual representation in real time.

- Basic Movement Mapping

This system converts the spatial movements of the dancer's torso and body into dynamic variables of visual texture. Before identifying unique morphemes, it displays rhythmic water wave patterns composed of basic lines, which are undulating. According to the rhythm and the movement of the X/Y axis, these textures will present a rhythmic and contracting effect. This is an emotional flow that emerges from physical movement and is continuously manifested in visual display and music.

TABLE IV. VISUAL MAPPING OF BASIC MOVEMENT

Parameter Type	Input Source	Mapping Logic	Visual Output
Beat Strength (Strong / Weak Beat)	pressure sensor	Strong beat → contraction (Noise: 0.5-1); Weak beat → relaxation(Noise: 0-0.5)	Pattern size
Y-axis (Vertical Movement)	Body position tracking	Higher position → higher brightness (Light⩾1); Lower position → darker tone (Light⩽1)	Pattern density
X-axis (Horizontal Movement)	Body position tracking	Larger motion →contraction (Noise: 0.5-1); smaller motion →relaxation(Noise: 0-0.5)	Pattern compression and release

- Morpheme Combination Mapping

Based on the combination of action morphemes, the system can determine and generate compound patterns with symbolic meanings. After identifying the trigger conditions of the "Three Bends", patterns will be extracted from the preset cultural symbol library. Combine them in real time and layer by layer to form a unified symbol symbolizing nature, life and ritual. The dynamics of these patterns change in accordance with the rhythm of the music, and the data from the pressure sensor is used again here. It creates a visual rhythm that is in sync with movements and sounds. When the beat is strong, the pattern becomes larger; when it is weak, it becomes smaller and lighter.

- Hand Gesture Mapping

By recognizing specific gesture morphemes, the system evokes visual symbols corresponding to the gestures. The "Peacock Hand" evokes the typical peacock pattern, imitating the soft light and shadow of feathers that spread out; The gesture of "leaf" creates the pattern of bamboo leaves and translucent, multi-layered bamboo forests. The gesture of "fish" imitates the swimming of a school of fish in the water, and the sense of fluidity and rhythm underwater is reflected. This gesture mapping provides a direct path from movement to image, allowing users to trigger and experience the emotional language of Dai dance during the dance process.

Fig. 2. Visual Mapping of Hand Gesture

E. System Implementation

- Software Platform and Communication

It employs a browser-based platform (p5.js + ml5.js) for real-time pose capture and recognition, together with an Arduino-based pressure-sensing module. Both motion and pressure data streams are transmitted as JSON to a local Node.js server via WebSocket and serial communication, respectively, and subsequently relayed in parallel to Max/MSP (via udpreceive) and TD (via CHOP). Max/MSP performs music synthesis, while TD handles visual rendering. Sharing the same low-latency communication pipeline, the two subsystems achieve real-time multimodal synchrony within a unified interaction framework. Graph Convolutional Networks (GCNs) were not used because the system targets real-time interactive performance, where controllability and low latency are essential. Rule-based mappings provide predictable behavior and lower computational cost compared to offline-trained GCN models.

Fig. 3. Software Platform and Communication Logic.

- Max/MSP Audio Generation

The generation of sounds is done in Max/MSP, based on the architecture of layered synthesis. The rhythm layer is generated by an 8-step beat matrix, with [metro] and [counter] being used for rhythmic control. [sfplay~] and [cycle~] objects are triggered by sampled and generated sounds such as elephant-foot drums, xylophones, and gongs, producing rhythmic patterns based on 4/2 time. The melody layer creates related tables of MIDI pitch mapping based on the Dai pentatonic scale. Body-space movement spatial data is translated by [makenote] and [noteout] objects into sample playback of the instruments of Bawo and Hulusi in the Kontakt sound engine, creating soft, flowing melodic lines. The system permits both random generation by [random] objects and preset triggering of the "wave-like motifs" in fixed form. Also, upon the identification of certain gestures such as the "Peacock Hand", the system triggers the related environment sound samples, such as bird songs and running water, via the WebSocket command. This generates a three-layer auditory scene of the Dai "human-nature symbiosis" by rhythm, melody, and sound effects.

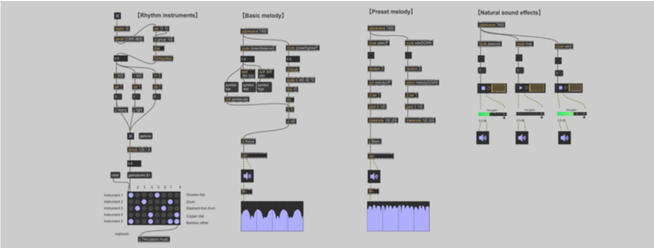

Fig. 4. Max/MSP Audio Generation.

- TouchDesigner Visual Generation

The visual module receives motion information via a WebSocket DAT in TD, which is subsequently processed through a CHOP network to generate continuous visual control signals. The overall visual system is structured into two major visual pipelines: (1) a baseline particle-flow graphics system that operates when no gesture is triggered, and (2) a layered motif system that activates upon gesture detection.

For the baseline particle-flow graphics, a SOP–TOP hybrid workflow is employed. A SOP [Tube]serves as the geometric foundation, which is repeatedly deformed using Twist SOP operators to generate curved and spiraling motion characteristics. The resulting geometry is rasterized into TOP space, where a [Transform] TOP combined with a [Noise] TOP introduces surface fluctuation. Specifically, fluctuations are produced by driving the translate tz parameter of the Noise TOP using the x-axis distance data extracted from the body position tracking module. Color variation in this pipeline is controlled by the y-axis body distance data. The incoming positional values are normalized to a -1 to 1 range via a [Math]CHOP, then exported to the [Constant]MAT parameters through CHOP Export, producing continuous color modulation in real-time.

The layered motif subsystem handles gesture-triggered visual patterns. Motion data received through WebSocket is binarized into trigger states (1 or 0). When a trigger state equals 1, a [Control] CHOP/TOP chain activates the corresponding visual motif, enabling motifs such as water-ripple sheets, bamboo-sway patterns, or peacock-feather overlays to appear. These motifs also inherit dynamic modulation driven by body tracking data; for example, positional deltas and rhythmic variation are applied to TOP transforms to synchronize motif motion with the performer's bodily rhythm.

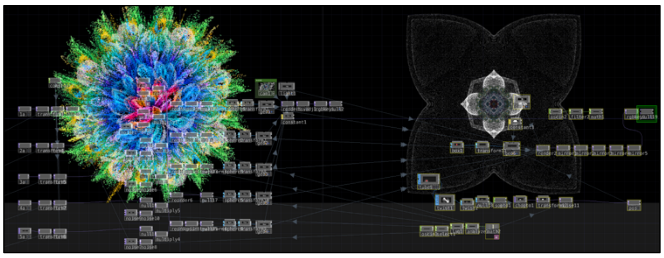

Fig. 5. TouchDesigner Visual Generation

IV. Design Experiment & User Experience

A. Experimental objective

The experiment combined two elements, technical functions and user experience, to evaluate the effectiveness of the system. The specific goals are as follows. (1) Check the real-time responses of movement, music and vision. (2) Determine whether the rhythm, melody and visual expression of the created music are in harmony. (3) Evaluate users' aesthetic experience, emotional response and cultural immersion. (4) Consider how embodied interaction can help people understand and participate in Dai culture.

B. Experimental Design

A total of 41 people participated in the experiment: 28 students majoring in art design, 5 middle-aged people (aged 31 to 52), 25 females and 16 males. None of the participants were students majoring in dance, but they had a basic understanding of art. To ensure the safety of motion capture and the accuracy of recognition, the experiment was conducted in a controlled environment of 4×4m.

This study is a low-risk one that does not cause harm to the human body and does not involve sensitive personal information or commercial interests. All participants signed an informed consent form before participating in the study, understanding the purpose, process and rights of the study. All data has been anonymized to ensure that personal identities cannot be traced.

The task is designed based on the typical dance movements of the Dai ethnic group. Participants will first simply learn the basic movements of the Dai ethnic group, then perform these combined movements in sequence (one set every 15 seconds), and then freely expand the movements based on real-time feedback. This process has three stages: (1) Familiarizing oneself with the movements; (2) Actual performance; (3) Free exploration.

This assessment combines technical objective measurements (frame rate, synchronization rate, delay) with perceptual information (interviews, questionnaires). The questionnaire adopts a Likert five-point scale (1 = strongly disagree, 5 = strongly agree). It covers three independent variables: system technical performance(X1), modal coordination(X2) and culturality question(X3).

We will record sensor data, identification results and music generation logs (at a sampling rate of 60 Hz), including displacement, pressure, pitch and visual trigger time difference. The subjective data were derived from 18 questionnaires, including 4 basic questions, 4 system technical performance questions, 3 modal coordination questions, 1 culturality question, and 6 questions related to the dependent variable of user experience. We also conducted semi-structured interviews, mainly focusing on physical participation and cultural understanding.

Data analysis includes: (1) Quantitative statistics of system latency and synchronization rate; (2) Measure the average value and standard deviation of the questionnaire; (3) Qualitative analysis of the interview content and summary of the theme, as well as evaluation of the system's performance in interactive experiences and cultural exchanges.

C. Experimental Results

- System Performance

This system has demonstrated good real-time performance in actual operation. After continuous testing, the average frame rate of the system reached 60 FPS, with a peak of 80.6 FPS and a minimum of 42.0 FPS. The frame loss rate has remained stable at 0.1%. In the motion jitter test, the average jitter value of the system was 3.0 pixels, indicating that the attitude tracking has high stability and accuracy. The motion-music response delay is 73 ms (SD=12), and the motion-visual delay is 126ms (SD=20), meeting the real-time requirements. The accuracy of action recognition varies with complexity. The recognition rate of high-amplitude postures such as "Three Bends" is the highest (96.68%), while the recognition rate of fine gestures such as "Leaf Hand" is slightly lower (88.7%).

- User Experience

User ratings show that the overall satisfaction rate of the system is 4.03. Among all dimensions, modal coordination (3.97) and culturality (4.12) scored relatively high. When using the system, users have increased their interest in Dai culture (4.22), and have a strong sense of presence (4.07) and artistic sense (4.00).

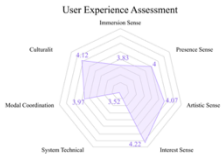

Fig. 6. Data of User Experience Assessment

Interview analysis revealed three themes. First, 70% of participants described experiencing their "body as a creative medium," generating music and visuals through movement. Second, users connected audio-visual feedback to Dai cultural symbols such as peacocks and flowing water. Third, real-time responsiveness enhanced sense of control and immersion.

A multiple linear regression analysis was conducted to examine how the three predictors influenced cultural experience perception. The model demonstrated high overall fit, $R^2 = 0.839$, indicating that the independent variables explained 83.9% of the variance. ANOVA showed that the regression model was statistically significant, $F(3, N–4) = 70.33$, $p < 0.001$, suggesting that the overall equation is valid (Table VI).

TABLE V. MODEL SUMMARY[B]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
	0.922[a]	0.851	0.839	0.2879837439	2.194

a. Predictors:(Constant) , X3,X2 X1

b. Dependent Variable: Y

TABLE VI. ANOVA[A]

Model	Sum of Squares	df	Mean Square	F	Sig.
Regression	17.499	3	5.833	70.333	<001 [b]
Residual	3.069	37	0.083		
Total	20.568	40			

c. Dependent Variable: Y

d. Predictors: (Constant), X3, X2. X1

Regression coefficients (Table VII) showed that motor coordination was the strongest predictor of system user experience perception ($\beta = 0.652$, $p < 0.001$), indicating that the system's motion–audio–visual synchronization performance plays a critical role.

TABLE VII. COEFFICIENTS[A]

	Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
Model	*B*	*Std. Error*	*Beta*	*t*	*Sig.*	*Tolerance*	*VIF*
(Constant)	0.083	0.280		0.297	0.768		
X1	0.201	0.090	0.219	2.231	0.032	0.419	2.386
X2	0.702	0.116	0.652	6.031	<0.001	0.345	2.897
X3	0.111	0.068	0.134	1.621	0.113	0.587	1.703

e. Dependent Variable: Y

The resulting regression equation based on unstandardized coefficients was:

$$Y=0.083+0.201X_1+0.702X_2+0.111X_3 \tag{4}$$

- Comprehensive Analysis

In this study, movement coordination emerged as the most significant predictor of experience, likely stemming from the system's inherent interactive nature. The core logic of our system is "dancer's movements to real-time music generation," where the amplitude, speed, and stability of movements directly determine the rhythm, melody, and harmonic changes in the sound. Consequently, the more users maintain continuous, smooth, and rhythmically synchronized movements, the more stable and structurally coherent the system-generated music becomes, naturally enhancing the immersive experience. Within this system, movement coordination transcends mere physicality to become the fundamental driver of musical generation quality. Furthermore, since participants are not professional dancers, they tend to form their experiential cognition more readily from the perspective of "Can I make the system produce the sounds I envision?" rather than through cultural imagery. Consequently, movement coordination assumes a dominant role.

Experiments confirm the system's technical capability to achieve effective synchronization of movement, music, and visuals. At the user experience level, embodied interaction has been validated to enhance cultural immersion. Users participate in sound-image generation through bodily movements, fostering a psychological experience of "dancing with culture" and demonstrating the applicability of embodied cognition in cultural experience design.

V. DISCUSSION

A. System Features and Advantages

The system proposed in this study achieves an effective balance among real-time responsiveness, musicality and cultural expression. Its core strength lies in treating bodily movements as the generative core of musical and visual narratives based on embodied cognition theory, thereby transcending traditional “input-output” interaction models. According to Varela, Thompson, and Rosch's Enactive Cognition Theory [33], cognition is not symbolic computation occurring in the brain but emerges through continuous coupling between body and environment. In our system, this theory manifests in two levels of embodied mechanisms:

- Cultural Activation through Kinesthetic Feedback

When users perform the “Three Bends” posture, the proprioception experienced—muscle tension changes, joint angle adjustments, center-of-mass shifts—forms cross-modal binding with the system-generated “wave theme” music. Research on embodied music cognition demonstrates that synchronized movement and sound activate shared neural representations, enhancing emotional engagement [34][35].In our experiment, the audio-visual imagery generated by the system through movement enabled 78% of participants to evoke natural imagery such as flowing water. This confirms that action-music cross-modal integration activated embodied understanding of Dai “water culture”.

- Bodily Internalization of Cultural Symbols

Lakoff and Johnson's Conceptual Metaphor Theory [36] posits that abstract concepts (e.g., “elegance,” “harmony”) originate from metaphorical mappings of bodily experiences. Our system constructs direct pathways between sensorimotor schemas and cultural symbols by mapping Dai dance's “Peacock Hand” to bird chirping sounds and “leaf hand” to bamboo forest imagery. Users no longer passively “view” peacock patterns but generate peacock symbols through precise hand control (thumb-index contact), and this active construction process promotes deep internalization of cultural meaning.

B. *Cultural and Creative Industries and Interdisciplinary Value*

This study presents a novel design approach for digital experiences of ethnic dance. Through embodied interaction, the system integrates users into real-time creation, facilitating a shift from “watching culture” to “experiencing culture” and even “recreating culture.” This expands the participatory and regenerative potential of traditional culture within contemporary contexts.

Methodologically, the thesis synthesizes computer vision, real-time audio-visual synthesis, cultural semiotics, and interaction design in order to create a cross-platform, expandable architecture that can be extended in all sorts of ways, such as in immersive exhibitions, digital museums, and cultural education, in order to promote the upgrading of the cultural content from static display to dynamic interaction. In a broader sense, the thesis demonstrates the shift of paradigm from “human-computer interaction” to “human-culture-media” cooperative creation, providing theoretical references as well as practical modes of the digital upgrading of the cultural and creative industries.

C. *Limitations and Future Work*

Despite the fact that the system shows stable overall performance, there are still some limitations. The motion recognition accuracy of the current system is limited by the visual algorithms of ml5.js, which can cause jitter or lost detections in the case of intricate movements. The periodic fluctuations in the visual rendering module under heavy load, combined with communication delay over the WS protocol, also affect the smoothness of multimodal synchronization. From a cultural standpoint, the system has limited vocabulary of Dai dances, and the scope of the cultural symbol generation dimension is quite narrow, which cannot accommodate in-depth and more extensive ethnic narratives.

In order to overcome these challenges, future research shall go in the following ways: Technologically, models such as Media Pipe Holistic shall be proposed in order to increase the motion recognition robustness. Culturally, it shall strive to construct an extensible “national dance culture morpheme database”. The database shall incorporate the body vocabulary, the rhythm patterns, and the visual symbols of various ethnic dances in order to create a computable cultural semantic system. The base shall accommodate interactive storytelling and emotional expressions, cross-culturally.

Conclusion

This study proposes and implements a multimodal “dance–music–visual” interaction system that takes Dai ethnic dance as its cultural prototype. The system adopts a dual-layer architecture that integrates body-structure control with motion-morpheme recognition, enabling real-time transformation of bodily movements into musical and visual outputs that embody ethnic aesthetic characteristics. The system thereby constructs an embodied form of digital cultural experience. Experimental results demonstrate that the proposed system ensures real-time interaction performance while effectively enhancing users’ cultural immersion and artistic engagement. The primary contribution of this work lies in introducing a digital pathway that shifts traditional dance from a mode of “performative viewing” to one of “embodied co-creation,” positioning bodily movement as a dynamic medium that bridges technological interaction and cultural narrative. This study provides a feasible paradigm for the digital preservation and innovation of ethnic cultural heritage, with strong potential for applications in interactive education, immersive exhibitions, and cultural and creative industries. Future work will focus on expanding the system’s adaptability across diverse dance genres and cultural contexts, further advancing multimodal interaction technologies within cultural heritage domains and fostering cross-disciplinary integration.

Acknowledgment

This study complies with the IEEE guidelines on artificial intelligence-assisted writing and data governance. ChatGPT is only used for language refinement. All the conceptual and experimental work was completed by the author.

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